CHAPTER 3

Vector calculus and the scalar product

> **Key point · Chapter path. Objectives.** Choose between projection, coordinates and angle; calculate a length; characterise orthogonality and a locus.
>
> **Prerequisites.** Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.
>
> **Yearly reference point.** 6–15 October: 8 hours

Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. For $A(1,2)$ and $B(4,-2)$, calculate $\overrightarrow{AB}$.
2. Calculate the norm of $(3,-4)$.
3. Simplify $\overrightarrow{AB}+\overrightarrow{BC}$.
4. Expand $(a+b)^2$.
5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Give its cosine.
6. What is the midpoint of $A(-2,1)$ and $B(4,3)$?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Compare, on the same problem, coordinates and vector calculus without a frame. Justify why your method uses the data most directly.

Investigate before formalising

Construct three points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for a segment AB of 6 cm. Conjecture the locus, then produce a proof that includes A and B.

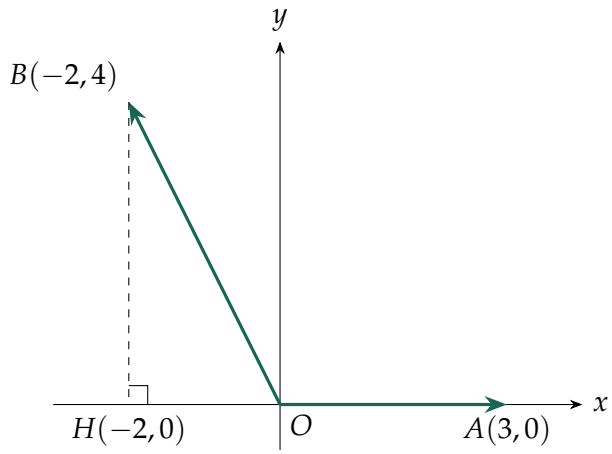
Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

3.1 Projection and angle

Key point · Definition and results. For two nonzero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their unoriented angle. With a zero vector, the product is 0. If H is the projection of B onto (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ equals $OA \times OH$ when H is on the ray $[OA)$, and its opposite otherwise.

Status and justification. Definition; the equivalence with the projection follows from the cosine in right triangles, with a negative algebraic projection for an obtuse angle.

Example 1. For two vectors of norms 3 and 4 forming 60°, the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120°, it is -6. If $A(3,0)$ and $B(-2,4)$, the projection is $H(-2,0)$ and the product is -6.



The projection H lies on the negative side of the horizontal axis. Hence $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Watch out. The result is a number, not a vector. A right angle corresponds to 0, not to 1.

Application. Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

3.2 Coordinates, norm and orthogonality

Key point · Definition and results. In an orthonormal basis, for $\vec{u} = (x, y)$ and $\vec{v} = (x', y')$, $\vec{u} \cdot \vec{v} = xx' + yy'$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For an orthonormal basis $(\vec{i}, \vec{j})$, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x'\vec{i} + y'\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Example 2. $\vec{u} = (2,3)$, $\vec{v} = (3,-2)$: product $6 - 6 = 0$. The two vectors are orthogonal. Their norm is $\sqrt{13}$. The product of $(2,3)$ with $(1,0)$ is 2: we recover the x-coordinate.

Watch out. The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

Application. For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal?

3.3 Vector calculus and identities

Key point · Definition and results. The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Hence $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. It follows that $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity accepted from the geometric definition; symmetry comes from the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Example 3. If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

Watch out. In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

Application. Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi and choosing a method

Key point · Definition and results. In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows us to calculate the third side from two sides and the included angle, or to find the cosine of an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2\overrightarrow{AB} \cdot \overrightarrow{AC}$; then introduce the expression with the cosine.

Example 4. $AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if one is allowed.

Watch out. The angle is included between the two sides of the product term. The minus sign is essential.

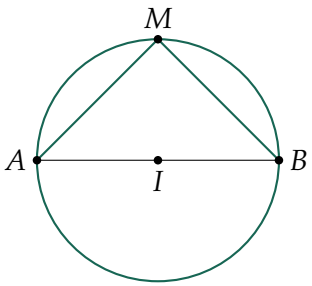
Application. A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

3.5 The circle with diameter AB

Key point · Definition and results. For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ is the circle with centre I and radius $\dfrac{AB}{2}$, including the points A and B.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel; what remains is $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, equality to zero is equivalent to the metric equation of the circle.

Example 5. $A(-2,0)$ and $B(2,0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0,1)$, the product is $1 - 4 = -3$; for $M(0,2)$, it is 0.



Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Watch out. The endpoints belong to the locus even though "right triangle" does not apply to a degenerate triangle.

Application. For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 3.1 · Direct calculations

In an orthonormal basis of the plane, consider the vectors $\vec{u} = (1, -2)$ and $\vec{v} = (4, 3)$.

1. Calculate the scalar product $\vec{u} \cdot \vec{v}$, then the norms $\|\vec{u}\|$ and $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u} + \vec{v}$ and deduce $\|\vec{u} + \vec{v}\|^2$.
3. Recover this last result using the identity involving $\vec{u} \cdot \vec{v}$.

Exercise 3.2 · A right triangle?

The plane is equipped with an orthonormal frame. Given the points $A(1, 1)$, $B(4, 2)$ and $C(0, 4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deduce that the triangle ABC is right-angled; state at which vertex.
3. Calculate the lengths of the two perpendicular sides, then the area of the triangle in area units.

Exercise 3.3 · Length without a frame

The quadrilateral $ABCD$ is a parallelogram with $AB = 4$, $AD = 3$ and $\widehat{BAD} = 60°$. The lengths are expressed in the same unit.

1. Draw a figure and calculate the scalar product $\overrightarrow{AB} \cdot \overrightarrow{AD}$.
2. Determine the exact lengths of the diagonals $[AC]$ and $[BD]$. Explain how the vectors corresponding to the diagonals are expressed in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. State which diagonal is longer and check that this result is consistent with your figure.

Exercise 3.4 · Recovering an angle

Two nonzero vectors $\vec{u}$ and $\vec{v}$ satisfy $\|\vec{u}\| = 2$, $\|\vec{v}\| = 6$ and $\vec{u} \cdot \vec{v} = -6$. Let θ be their unoriented angle, between $0°$ and $180°$.

1. Determine the exact value of $\cos\theta$, then the measure of θ.
2. Explain how the sign of the scalar product made it possible to predict that the angle would be obtuse.

Exercise 3.5 · A variable locus

The points A and B are distinct and fixed. Let I be the midpoint of $[AB]$. For a fixed real number k, we look for the set of points M of the plane that satisfy $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product in terms of MI and AB using Chasles's relation and a scalar identity.
2. Describe precisely the set sought according to the value of k. For each case, state whether it is empty, a single point or a circle; then give its characteristics.
3. Explain the role of the fact that a squared distance is positive or zero.

Exercise 3.6 · Work done by a force

An object moves in a straight line from A to B over a distance of 4 m. It is subject to a constant force $\vec{F}$ of magnitude 50 N. The work done by this force is defined by $W = \vec{F} \cdot \overrightarrow{AB}$ and is expressed in joules.

1. Calculate W when the angle between $\vec{F}$ and $\overrightarrow{AB}$ is $60°$.
2. Repeat the calculation for an angle of $120°$.
3. Compare the signs of the two works and explain, in each situation, whether the force helps the displacement or opposes it.

Exercise 3.7 · Contrapositive

We wish to prove the following statement, valid for two vectors of the plane:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal.}$$

1. State the contrapositive of this implication.
2. Prove this contrapositive using a scalar identity, then conclude about the initial statement.

Exercise 3.8 · Delayed review

A triangle ABC satisfies $AB = 3$, $AC = 4$ and $BC = 5$. We look for $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using coordinates.

1. Calculate this scalar product using a formula involving the three lengths.
2. Recover the result by determining the nature of the triangle. Name the theorem used and its numerical hypothesis.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate the scalar product in the orthonormal basis and detail the two products of coordinates.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. Use in turn the coordinates, Chasles's relation and the fact that I is the midpoint.

MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $(2,1)\cdot(3,-4)$ equals
   - A. 2
   - B. 10
   - C. $(6,-4)$
   - D. -2
2. The norm of $(-3,4)$ equals
   - A. 7
   - B. 25
   - C. 5
   - D. 1
3. Two vectors with zero product are
   - A. equal
   - B. collinear
   - C. opposite
   - D. orthogonal
4. Norms 2 and 3, angle 60°: product
   - A. 6
   - B. 3
   - C. $\sqrt{3}$
   - D. 0
5. $\|u-v\|^2$ equals
   - A. $\|u\|^2-\|v\|^2$
   - B. $\|u\|^2+\|v\|^2$
   - C. $\|u\|^2-2u\cdot v+\|v\|^2$
   - D. $\|u\|^2+2u\cdot v+\|v\|^2$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. An obtuse angle between nonzero vectors gives a product that is
   A. positive
   B. zero
   C. equal to 1
   D. negative
2. In an orthonormal basis, $u \cdot i$ is
   A. the x-coordinate of u
   B. the norm of u
   C. the y-coordinate of u
   D. always zero
3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is
   A. the perpendicular bisector
   B. the line AB
   C. the circle with diameter AB
   D. the midpoint alone
4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ equals
   A. -3
   B. -6
   C. 6
   D. 9
5. For $AB = AC = 2$ and a right angle at A, BC equals
   A. 4
   B. 2
   C. 8
   D. $2\sqrt{2}$

True or false? Justify.

1. The scalar product of two vectors is always positive.
2. The zero vector is orthogonal to every vector.
3. $u \cdot v = 0$ implies that one of the vectors is zero.

> **Key point · Self-assessment.** I distinguish the norm from its square; I choose a suitable formula; I check the frame; I write an equivalence for a locus.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.